POOJAA V K **B020**

Detecting Financial Shocks & Segmenting Behavioral Risk Profiles using Unsupervised Learning

UNSUPERVISED MACHINE LEARNING PROJECT

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1. Abstract

This project presents an unsupervised learning framework for detecting financial shocks and segmenting transaction behavior into meaningful risk profiles. In the absence of labeled fraud data, anomaly detection techniques such as Isolation Forest and DBSCAN are employed to identify unusual transaction patterns. Clustering methods, including K-Means and Gaussian Mixture Models, are further used to group transactions based on behavioral similarities.

A composite Financial Stress Index (FSI) is developed to quantify transaction-level risk by integrating multiple engineered features. The results demonstrate that unsupervised learning can effectively uncover hidden structures in financial data and support proactive risk assessment. The final outputs include enriched datasets, cluster summaries, and interpretable behavioral segments, making the solution suitable for real-world financial monitoring systems.

2. Introduction

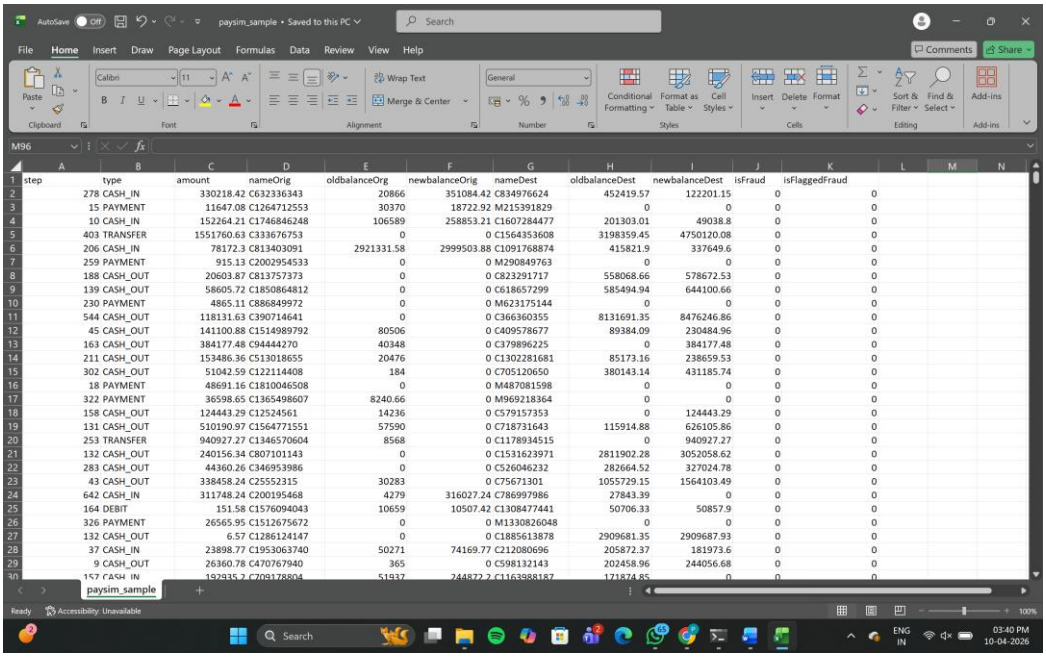
Financial fraud and systemic risks often manifest as subtle anomalies within large volumes of transactional data. Traditional supervised learning techniques depend on labeled datasets, which are frequently unavailable or incomplete in real-world financial environments. This limitation necessitates the use of unsupervised learning methods capable of discovering hidden patterns without prior labels.

The objective of this project is to develop a comprehensive pipeline that detects anomalous transactions, segments behavioral patterns, and quantifies financial risk. The approach integrates anomaly detection, clustering, and risk scoring into a unified framework, enabling a shift from reactive fraud detection to proactive risk monitoring.

3. Dataset Description

The dataset used in this study is the PaySim dataset, a synthetic dataset designed to simulate real-world mobile money transactions. It contains approximately 100,000 transactions with features such as transaction type, transaction amount, and account balances before and after each transaction.

The dataset is highly imbalanced, with fraudulent transactions comprising approximately 0.14% of the total records. Additionally, the dataset contains no missing values, making it suitable for direct analysis. However, the absence of strong correlations between individual features and fraud highlights the need for advanced feature engineering.



step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud
278	CASH_IN	330218.42	C632336343	20866	351084.42	C834976624	452419.57	122201.15	0	0
15	PAYMENT	11647.08	C1264712553	30370	18722.92	M215391829	0	0	0	0
10	CASH_IN	152264.21	C1746846248	106589	258853.21	C1607284477	201303.01	49038.8	0	0
403	TRANSFER	1551760.63	C333676753	0	0	C1564353608	3198359.45	4750120.08	0	0
206	CASH_IN	78172.3	C813403091	2921331.58	2999503.88	C1091768874	415821.9	337649.6	0	0
259	PAYMENT	915.13	C2002954533	0	0	M290849763	0	0	0	0
188	CASH_OUT	20603.87	C813757373	0	0	C823291717	558068.66	578672.53	0	0
139	CASH_OUT	58605.72	C1850864812	0	0	C618657299	585494.94	644100.66	0	0
230	PAYMENT	4865.11	C886849972	0	0	M623175144	0	0	0	0
544	CASH_OUT	118131.63	C390714641	0	0	C366360355	8131691.35	8476246.86	0	0
45	CASH_OUT	141100.88	C1514989792	80506	0	C409578677	89384.09	230484.96	0	0
163	CASH_OUT	384177.48	C944444270	40348	0	C379896225	0	384177.48	0	0
211	CASH_OUT	153486.36	C513018655	20476	0	C1302281681	85173.16	238659.53	0	0
302	CASH_OUT	51042.59	C122114408	184	0	C705120650	380143.14	431185.74	0	0
18	PAYMENT	48691.16	C1810046508	0	0	M487081598	0	0	0	0
322	PAYMENT	36598.65	C1365498607	8240.66	0	M969218364	0	0	0	0
158	CASH_OUT	124443.29	C12524561	14236	0	C579157353	0	124443.29	0	0
131	CASH_OUT	510190.97	C1564771551	57590	0	C718731643	115914.88	626105.86	0	0
253	TRANSFER	940927.27	C1346570604	8568	0	C1178934515	0	940927.27	0	0
132	CASH_OUT	240156.34	C807101143	0	0	C1531623971	2811902.28	3052058.62	0	0
283	CASH_OUT	44360.26	C346953986	0	0	C526046232	282664.52	327024.78	0	0
43	CASH_OUT	338458.24	C25552315	30283	0	C75671301	1055729.15	1564103.49	0	0
642	CASH_IN	311748.24	C200195468	4279	316027.24	C786997986	27843.39	0	0	0
164	DEBIT	151.58	C1576094043	10659	10507.42	C1308477441	50706.33	50857.9	0	0
326	PAYMENT	26565.95	C1512675672	0	0	M1330826048	0	0	0	0
132	CASH_OUT	6.57	C1286124147	0	0	C1885613878	2909681.35	2909687.93	0	0
37	CASH_IN	23898.77	C1953063740	50271	74169.77	C212080696	205872.37	181973.6	0	0
9	CASH_OUT	26360.78	C470767940	365	0	C598132143	202458.96	244056.68	0	0
157	CASH_IN	192935.2	C709178804	51937	244872.2	C1163988187	171874.85	0	0	0

Figure 1 Dataset Overview

4. Data Understanding

Exploratory data analysis was conducted to understand the distribution and characteristics of the dataset. The analysis revealed that transaction amounts are highly skewed, necessitating a logarithmic transformation to stabilize variance and improve model performance. Transaction type analysis showed that CASH_OUT and PAYMENT dominate the dataset, while TRANSFER transactions exhibit relatively higher fraud rates. Temporal analysis indicated that fraudulent transactions are distributed across the timeline rather than concentrated in specific intervals. Furthermore, correlation analysis demonstrated that no single feature strongly predicts fraud, reinforcing the need for multi-dimensional feature engineering.

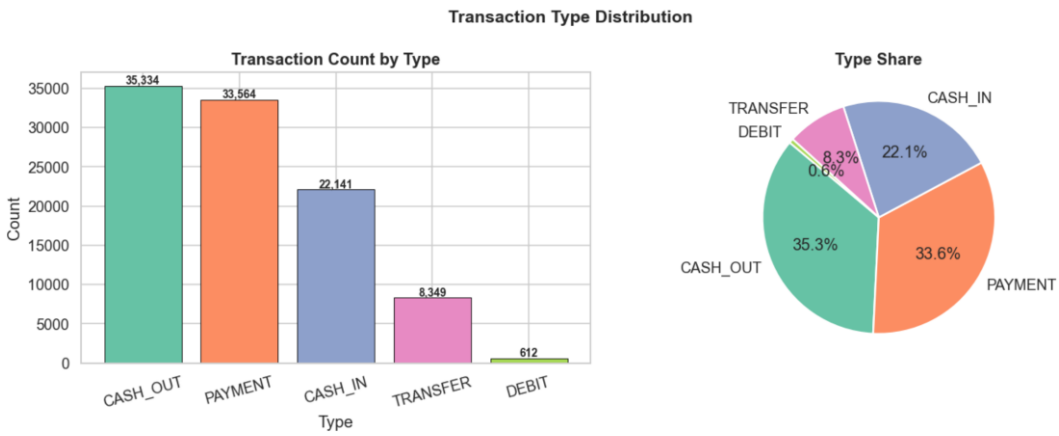


Figure 2 Transaction Type Distribution

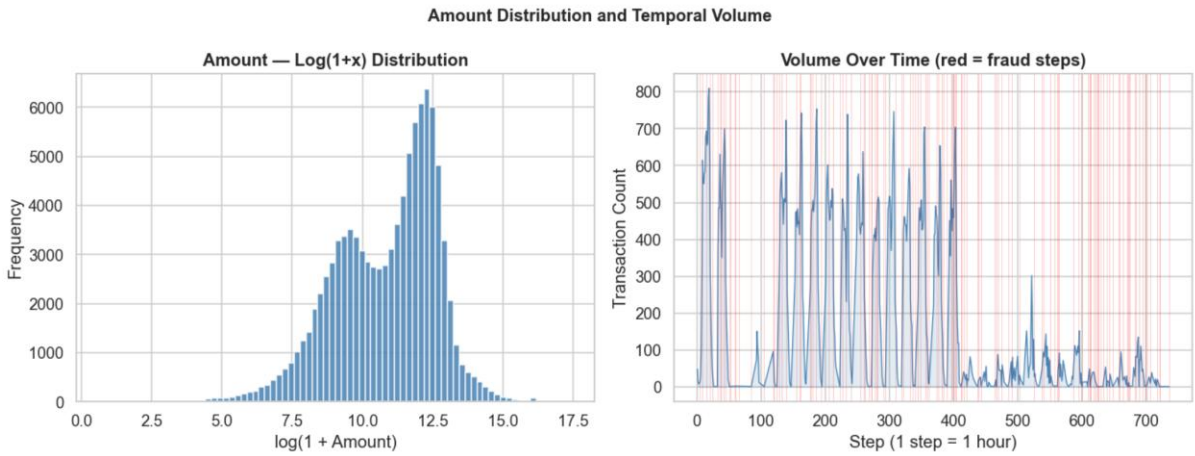


Figure 3 Amount Distribution (Log Scale) and Temporal Trend

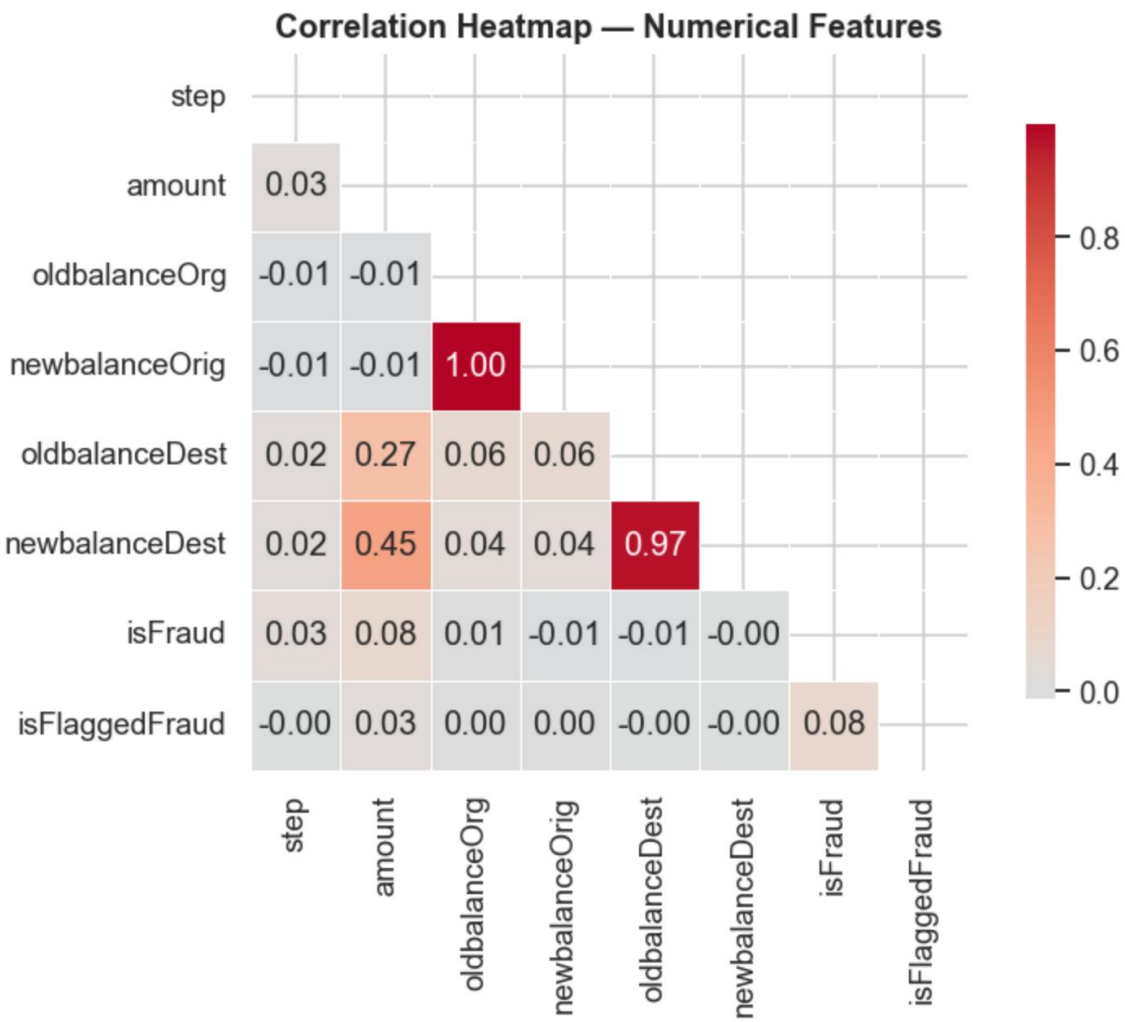


Figure 4 Correlation Heatmap of Numerical Features

5. Methodology

5.1 Feature Engineering

Due to the absence of repeated user-level transactions, feature engineering was performed at the transaction level to capture financial behavior. Several derived features were created to represent balance dynamics, transactional intensity, and anomaly indicators. Balance-related features such as balance drain and balance change ratio capture sudden depletion of funds. Ratio-based features such as amount-to-balance provide contextual insight into transaction magnitude. Additionally, binary indicators such as full drain, no destination credit, and amount mismatch were introduced to explicitly encode suspicious transaction patterns. These engineered features significantly enhance the model's ability to detect anomalies and form the foundation for subsequent analysis.

	log_amount	balance_drain	balance_change_ratio	amount_to_orig_bal	full_drain	no_dest_credit	amount_mismatch	hour
0	11.790856	0.00	0.000000	132038.410000	0	0	1	1
1	7.509659	1824.59	0.072580	0.072580	0	1	0	1
2	7.093770	1203.44	0.040192	0.040192	0	1	0	1
3	6.906094	997.34	0.009601	0.009601	0	1	0	1
4	8.523308	0.00	0.000000	5029.670000	0	1	1	1

Figure 5 Engineered Features Sample

5.2 Anomaly Detection

Anomaly detection was performed using two complementary unsupervised techniques: Isolation Forest and DBSCAN. Isolation Forest identifies anomalies by isolating data points through random partitioning, making it highly effective for high-dimensional datasets. It was configured to flag approximately 5% of transactions as anomalous. DBSCAN, a density-based clustering algorithm, identifies anomalies as points that do not belong to any dense region. Due to its computational complexity, it was applied to a sampled subset of the data. To improve robustness, the outputs of both models were combined using a union approach. This ensures that anomalies detected by either method are retained, thereby improving overall detection coverage.

```
Shocks flagged : 5,000  (5.00% of transactions)
```

Figure 6 Isolation Forest Shock Detection Output

```
Shocks flagged in sample : 148  (0.15% of total)
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Figure 7 DBSCAN Anomaly Detection Output

5.3 Clustering and Segmentation

Clustering techniques were applied to group transactions into behavioral segments. Both K-Means and Gaussian Mixture Models were evaluated using metrics such as the Silhouette Score and Davies-Bouldin Index. K-Means was selected as the final model due to its better performance and interpretability. The optimal number of clusters was determined to be four, ensuring meaningful segmentation while maintaining model simplicity.

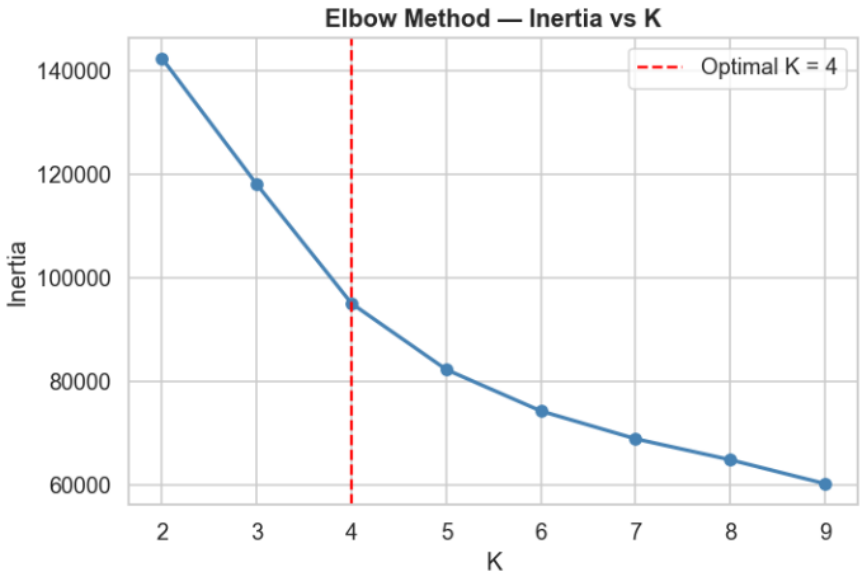

Figure 8 Elbow Method for Optimal K Selection

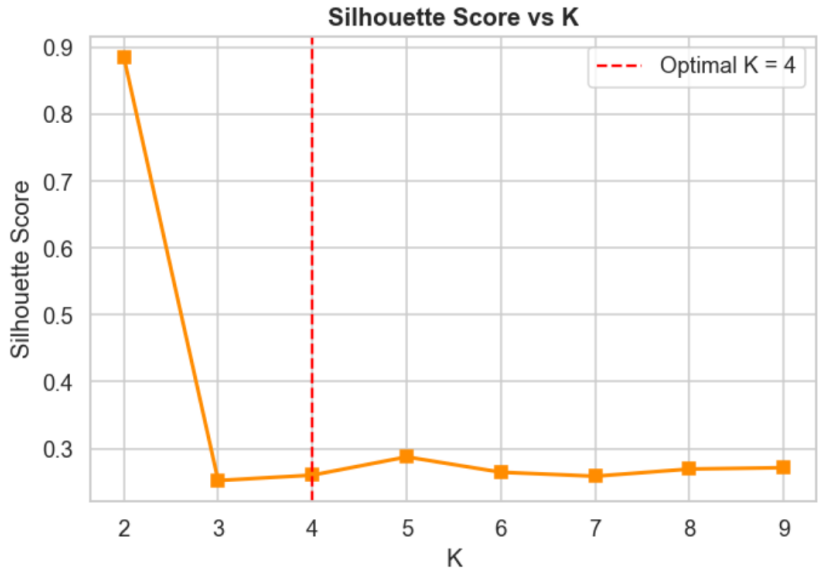

Figure 9 Silhouette Score vs Number of Clusters

6. Dimensionality Reduction and Visualization

To visualize high-dimensional data, dimensionality reduction techniques such as Principal Component Analysis (PCA) and t-SNE were applied. PCA provided a linear projection of the data, while t-SNE captured non-linear relationships.

The visualizations confirmed clear separation between clusters and demonstrated that anomalous transactions occupy distinct regions in the feature space.

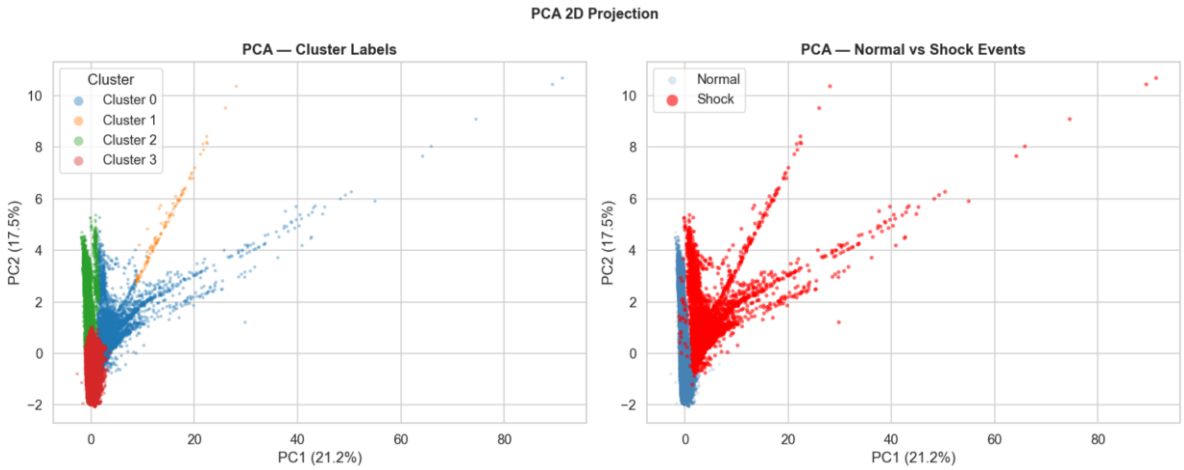

Figure 10 PCA Projection of Clusters

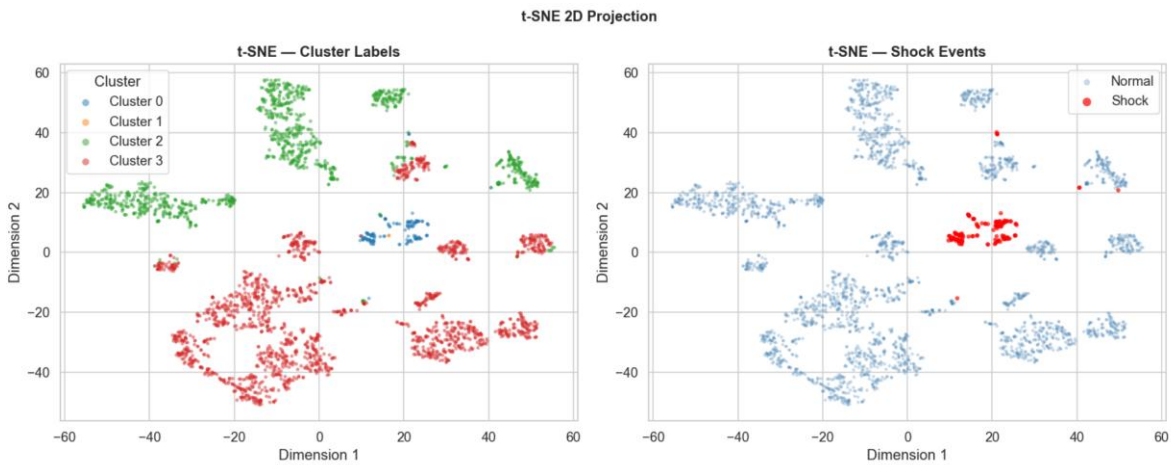

Figure 11 t-SNE Visualization of Clusters and Shock Events

7. Evaluation Metrics

Algorithm	Silhouette Score	Davies-Bouldin Index	Best Use Case
K-Means (K=4)	0.2822	1.1851	Hard clustering with clear, interpretable segments
GMM (K=4)	0.2733	1.5393	Soft clustering with probabilistic assignments
DBSCAN	N/A	N/A	Density-based anomaly detection

K-Means is selected as the final clustering model due to its superior performance in terms of cluster separation and compactness, as indicated by a higher Silhouette Score and lower Davies-Bouldin Index compared to GMM.

8. Results and Discussion

The results of the analysis demonstrate the effectiveness of the proposed unsupervised learning framework in detecting financial shocks and segmenting transaction behavior into meaningful risk profiles.

8.1 Cluster-Level Analysis

The cluster-wise analysis reveals significant variation in risk characteristics across segments. The High Risk cluster exhibits an extremely high shock rate (≈0.998), along with a comparatively higher fraud rate (≈2.69%) and the highest average Financial Stress Index (FSI ≈0.93), clearly identifying it as the most critical segment.

In contrast, the Stable cluster shows negligible fraud occurrence and minimal balance disruption, representing normal transactional behavior. The Shock-Prone cluster shows moderate anomaly presence with a shock rate of approximately 0.012, while the Moderate Risk cluster exhibits very low anomaly occurrence (≈0.001), indicating relatively stable but occasionally fluctuating behavior.

These results confirm that the clustering process successfully separates transactions based on risk intensity and behavioral characteristics.

	cluster_label	segment	n_transactions	shock_rate	avg_shock_magnitude	full_drain_rate	fraud_rate	avg_FSI
0	0	High Risk	4432	0.9984	2.7755	0.3360	0.0269	0.9310
1	1	Stable	113	1.0000	5.0875	0.0000	0.0000	0.7812
2	2	Shock-Prone	35097	0.0127	0.0249	0.1361	0.0003	0.4789
3	3	Moderate Risk	60358	0.0010	0.0017	0.2939	0.0002	0.5164

Figure 12 Cluster-wise Risk Metrics including Shock Rate, Fraud Rate, and FSI

8.2 Behavioral Segmentation

The distribution of transactions across segments indicates that the majority belong to the Moderate Risk and Shock-Prone categories, while High Risk transactions constitute a very small proportion. This reflects real-world financial systems where high-risk events are rare but highly impactful.

Further analysis of cluster profiles shows that High Risk segments exhibit significantly higher balance drain, shock magnitude, and full drain rates compared to other clusters. In contrast, Stable segments maintain consistently low values across all risk indicators.

This demonstrates that the segmentation is meaningful and captures real behavioral differences among transactions.

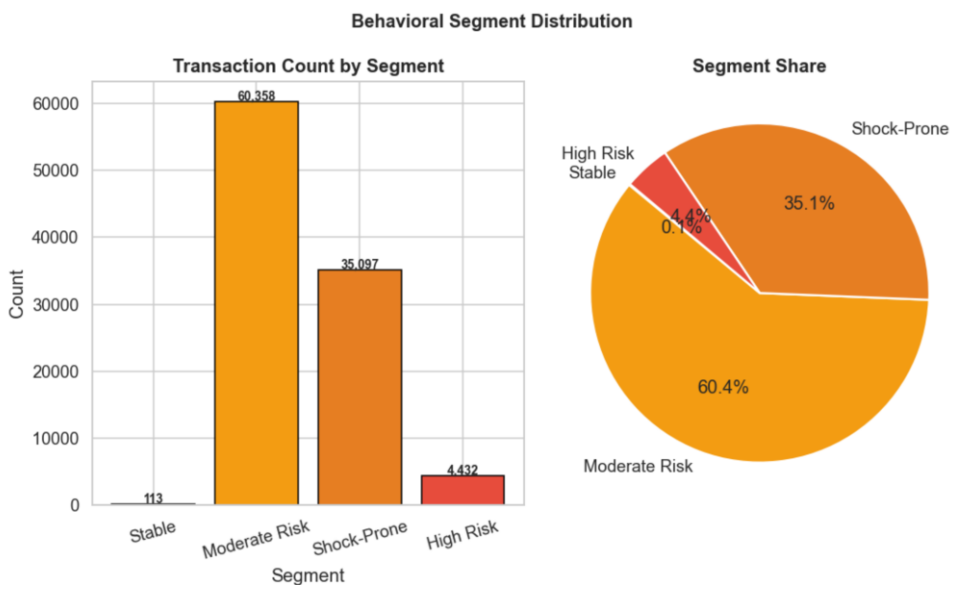

Figure 13 Segment Distribution

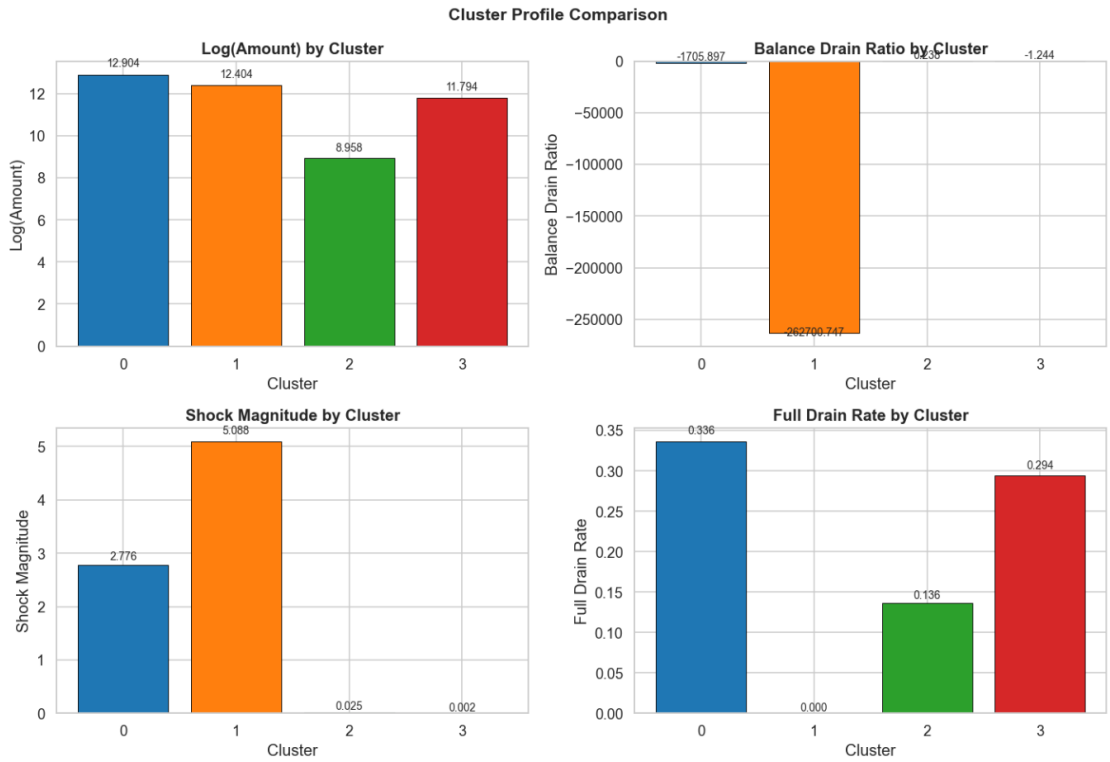

Figure 14 Cluster Profile Comparison

8.3 Financial Stress Index (FSI) Analysis

The Financial Stress Index distribution indicates a well-defined spread of risk scores across transactions. The segmentation into Low, Medium, and High risk tiers is balanced, with each category containing approximately one-third of the dataset.

Cluster-wise comparison shows that High Risk clusters consistently exhibit higher FSI values, while Stable clusters maintain lower scores. This alignment confirms that the FSI effectively captures transaction-level risk in a continuous manner.

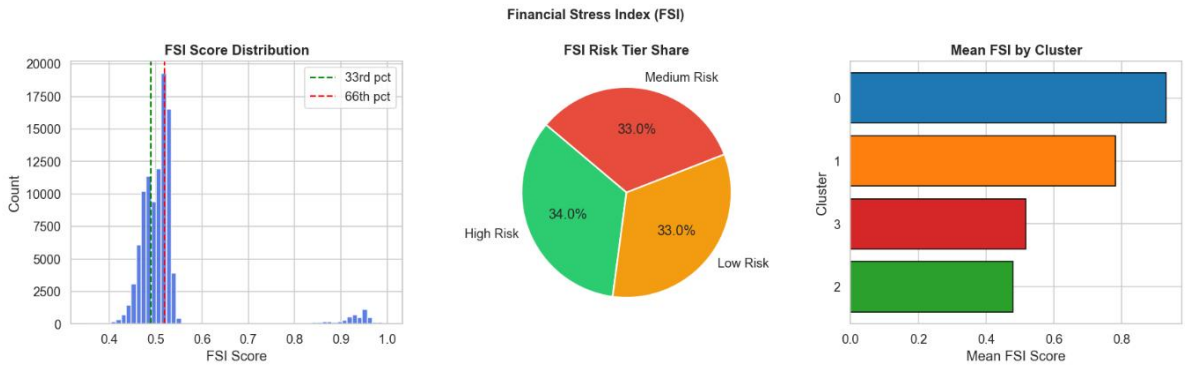

Figure 15 Distribution of Financial Stress Index and Cluster-wise Comparison

8.4 FSI Validation

The correlation analysis demonstrates a strong positive relationship between FSI and key risk indicators such as shock events and shock magnitude. This confirms that the index accurately reflects underlying anomaly patterns.

Additionally, the comparison across FSI tiers shows a clear monotonic increase in risk indicators. High-risk transactions exhibit significantly higher shock rates (≈0.148), full drain rates (≈0.35), and fraud occurrence (≈0.0039) compared to lower tiers.

This validates the effectiveness of FSI as a robust composite risk metric.

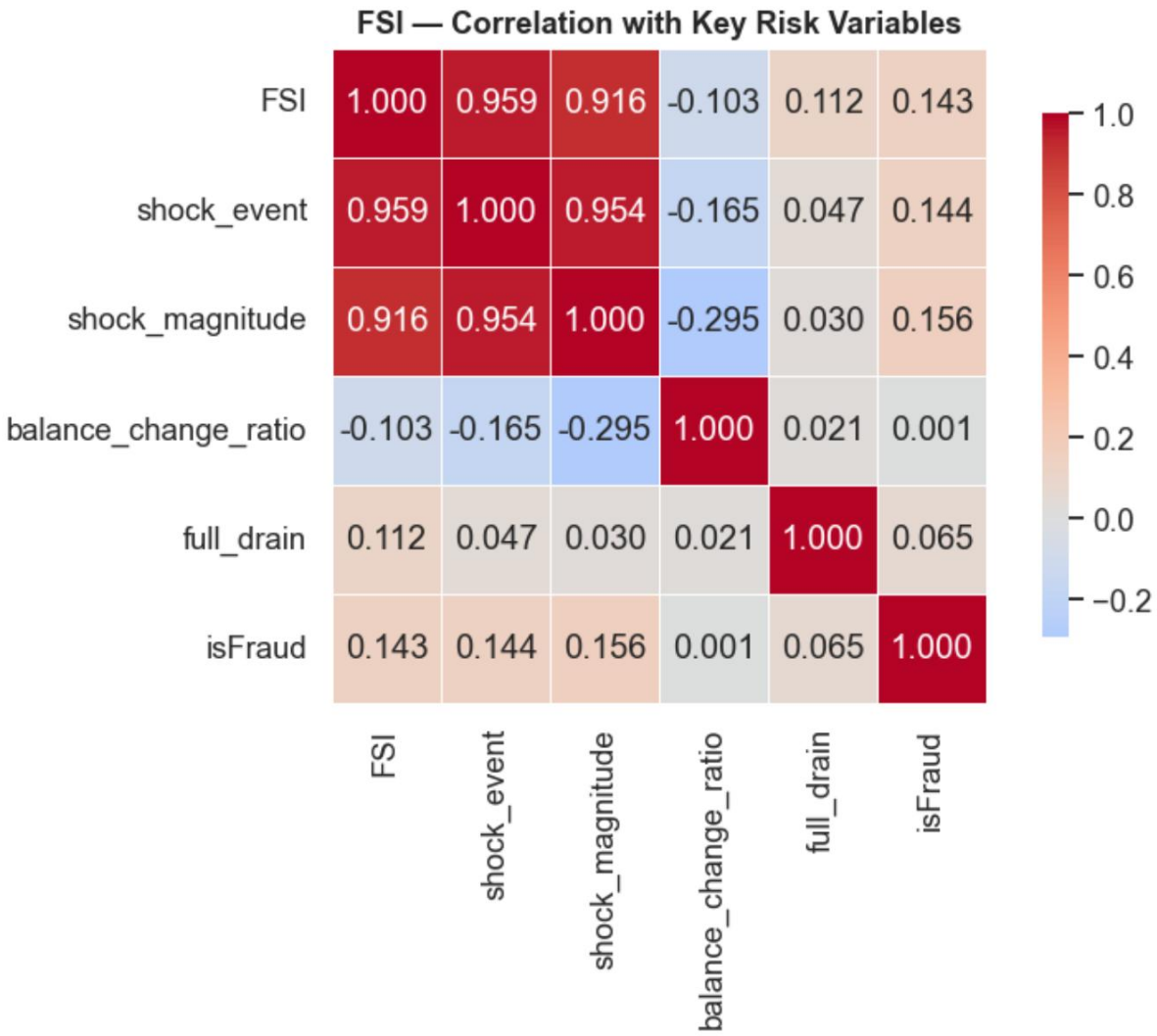

Figure 16 FSI Correlation Heatmap

fsi_tier	shock_event	balance_change_ratio	full_drain	isFraud
Low Risk	0.0000	0.1266	0.1097	0.0001
Medium Risk	0.0000	-0.8158	0.2543	0.0002
High Risk	0.1483	-1096.7560	0.3526	0.0039

Figure 17 Risk Indicators Across FSI Tiers

8.5 Anomaly Detection Effectiveness

The comparison of anomaly detection models shows that Isolation Forest identifies the majority of anomalies (≈4894 transactions), while DBSCAN detects a smaller but distinct subset (≈42 transactions). The overlap between the two models is minimal (≈106 transactions), resulting in a very low Jaccard similarity score (~0.02).

This indicates that the two models capture different types of anomalous behavior. Their combination improves overall detection coverage and reduces the likelihood of missing critical financial shocks.

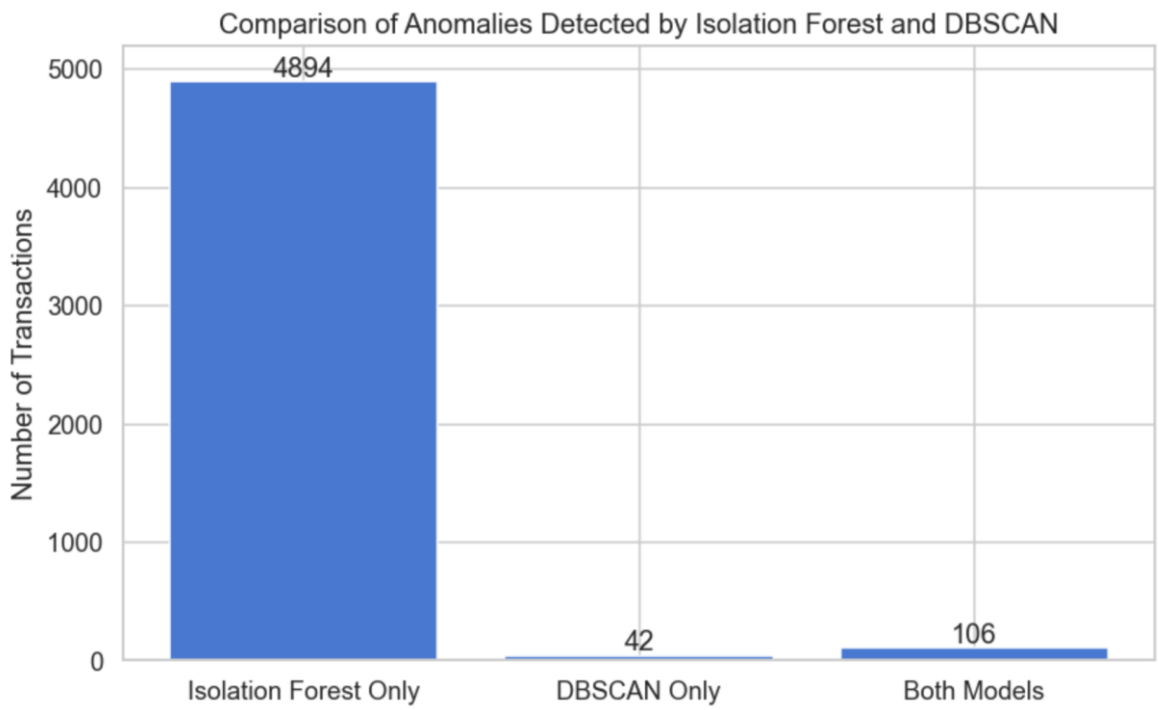

Figure 18 Comparison of Anomalies Detected by Isolation Forest and DBSCAN

8.6 Temporal Analysis of Shock Events

The temporal analysis shows that shock events are distributed across the entire timeline rather than being concentrated in specific intervals. This indicates that anomalies are not solely dependent on transaction volume or time-based patterns.

Instead, financial shocks arise due to underlying transactional characteristics, reinforcing the importance of feature-driven anomaly detection methods.

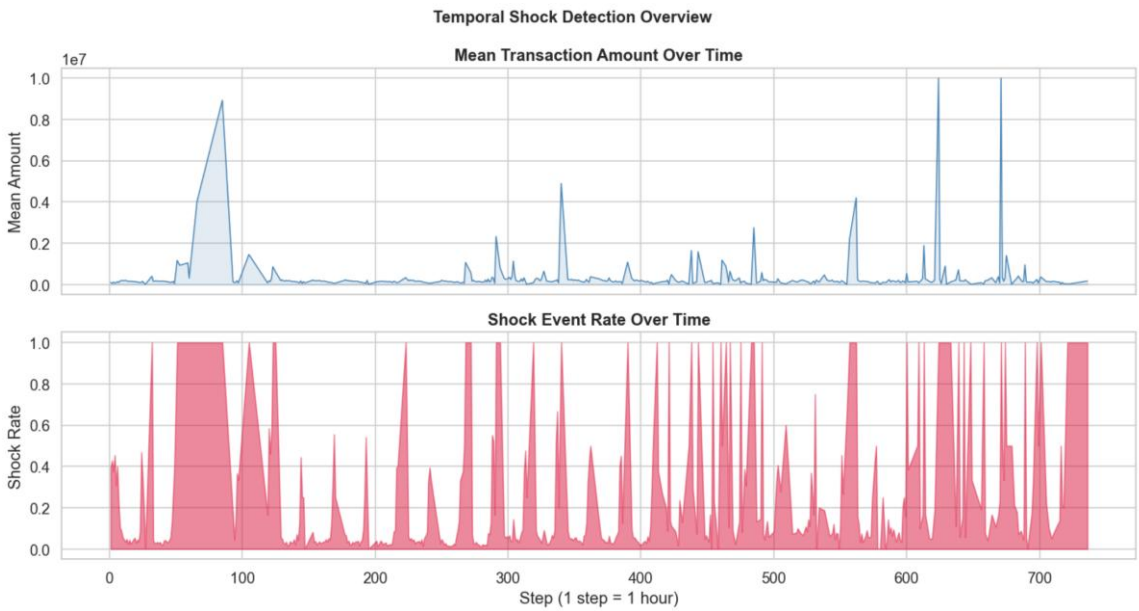

Figure 19 Temporal Distribution of Shock Events

9. Key Findings

The analysis reveals several important insights:

- The High Risk segment exhibits near-total anomaly concentration, with shock rates approaching 100%, confirming the effectiveness of anomaly detection methods.
- The Financial Stress Index (FSI) shows a strong monotonic relationship with key risk indicators, validating it as a reliable composite risk metric.
- Isolation Forest detects the majority of anomalies, while DBSCAN identifies a smaller but distinct subset, demonstrating complementary detection behavior.
- Clustering successfully segments transactions into meaningful behavioral groups, with clear separation between stable and high-risk patterns.
- Shock events are distributed across time, indicating that anomalies are driven by transactional characteristics rather than temporal trends.

10. Conclusion

This project successfully developed an unsupervised learning pipeline for detecting financial shocks and segmenting transaction behavior. By integrating anomaly detection, clustering, and risk scoring, the study provides a comprehensive approach to financial risk assessment.

The results validate the effectiveness of unsupervised learning in identifying hidden patterns and generating actionable insights. The proposed system is scalable and can be integrated into real-world financial monitoring and fraud detection systems. The integration of anomaly detection, clustering, and composite risk scoring provides a scalable and interpretable framework for real-world financial risk monitoring systems.

11. Business Applications

The proposed framework can be applied in real-world financial systems for fraud detection and risk monitoring. The anomaly detection models help identify suspicious transactions in real time, enabling early intervention. The Financial Stress Index (FSI) provides a continuous risk score, allowing institutions to prioritize high-risk transactions efficiently.

Additionally, the behavioral segmentation of transactions into risk personas supports customer profiling and risk-based monitoring. Financial institutions can use these insights for targeted analysis, improved decision-making, and regulatory compliance, particularly in anti-money laundering (AML) processes.

12. Limitations

The study is based on a synthetic dataset, which may not fully capture real-world financial complexities and behavioral variations. Additionally, the absence of labeled data limits the ability to directly evaluate the accuracy of anomaly detection models.

Furthermore, the use of sampling for DBSCAN due to computational constraints may lead to partial detection of density-based anomalies. The clustering approach assumes a fixed number of segments, which may not fully represent the underlying complexity of transaction behavior.

13. Future Work

Future enhancements may include the integration of real-world labeled datasets to improve model accuracy through hybrid approaches. Advanced techniques such as deep learning-based anomaly detection can be explored to capture complex patterns. The system can also be extended to real-time data processing for continuous monitoring. Further improvements in feature engineering and the incorporation of explainable AI techniques can enhance interpretability and practical applicability.

14. References

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